

NAYYAB FATIMA ALI

Toronto, ON | (647) 929-5446 | nayyab.ali@torontomu.ca | [LinkedIn](#)

SUMMARY OF QUALIFICATIONS

- Computer Engineering student at Toronto Metropolitan University with hands-on experience in embedded software, vehicle controls, hardware-software integration, and engineering software development using C, C++, Python, Java, MATLAB, and JavaScript
- Senior Designer with Toronto Metropolitan Formula Racing's Electric Powertrain team, contributing to VCU development, CAN communication, dashboard and driver-interface systems, sensor integration, battery State of Charge estimation, and low-voltage electrical systems
- Experience with embedded platforms including STM32, ESP32, Teensy, and Arduino for communication testing, prototyping, sensor integration, and real-time vehicle applications
- Junior Software Engineer experience with MACH, contributing to GUI refactoring, debugging, documentation, and maintainability for a rocket-engine control interface using React, Python, Git, and GitHub
- Familiar with CAN, UART, SPI, I²C, VHDL, digital logic, circuit analysis, PCB design tools, oscilloscopes, function generators, and electrical integration
- Interested in embedded software, firmware, vehicle control systems, EV architecture, battery and energy management, automated testing, and reliable real-time systems

EDUCATION

Bachelor of Computer Engineering (B.Eng)

Sept 2023 - Present

Toronto Metropolitan University | Toronto, ON

- **Relevant Coursework:** Circuit Analysis, Object-Oriented Programming (Java/C/C++), Data Structures & Algorithms, Digital Systems (VHDL), Electric Networks
- **Relevant Extracurriculars:** Metropolitan Undergraduate Engineering Society External Committee Member and IEEE Computer Society TMU Chapter Co-Chair

RELEVANT EXPERIENCE

Senior Designer Electrical Powertrain

Sept 2025 - Present

Toronto Metropolitan Formula Racing

- Develop embedded software and vehicle-control systems for a Formula SAE electric race car as a Senior Designer within the Electric Powertrain subsystem
- Lead development and updates of the vehicle dashboard and driver-interface architecture, integrating real-time vehicle data into a multi-state display for driving, diagnostics, fault monitoring, and endurance energy management
- Develop and update dashboard firmware using Teensy microcontrollers, UART communication, CAN data structures, and embedded C/C++ to process and display vehicle parameters including speed, battery State of Charge, temperatures, warnings, and drivetrain status
- Contribute to Vehicle Control Unit development and testing using STM32-based embedded systems, supporting vehicle-state logic, communication interfaces, and hardware-software integration
- Developed a battery State of Charge estimation implementation in C using pack-voltage measurements, voltage-divider scaling, lookup tables, and interpolation for real-time driver and vehicle monitoring
- Support CAN-based communication between vehicle subsystems including the VCU, motor controller, BMS, sensors, and dashboard, contributing to system-level data flow and debugging
- Design, assemble, and validate low-voltage electrical systems and wiring harnesses through soldering, crimping, sensor installation, accumulator preparation, continuity testing, and electrical troubleshooting

Junior Software Engineer

Sept 2024 - Apr 2025

Metropolitan Aerospace Combustion Hub (MACH)

- Refactored, cleaned, and documented a Graphical User Interface (GUI) codebase for rocket engine control, improving readability, maintainability, and onboarding support for future developers
- Collaborated with software and hardware leads using Git/GitHub version control to track changes, review updates, and support integration of GUI functionality with system requirements
- Supported development of engineering software tools using React and Python, gaining experience with code organization, debugging, documentation, and cross-functional aerospace software workflows